ContextBridge — Capability Roadmap

Target capabilities and implementation checklist

ContextBridge's target scope is more than a filesystem tool. It covers remote computer access, terminal execution, file editing, document handling, process management, search, device management, and MCP integration.

Sections 1–8 describe the target capabilities by area. Section 9 is the implementation checklist, and section 10 covers synchronization.

1. Filesystem operations

Capability	Description
Read a file	Read individual files
Read multiple files	Read several files in one operation
Write a file	Create, overwrite, or append to files
Edit a file	Make targeted edits without rewriting an entire file
Create a directory	Create folders and nested directory structures
List a directory	List files and folders recursively
Move a file	Move or rename files and directories
File information	Retrieve size, dates, permissions, type, and other metadata

Additional capabilities include reading selected line ranges, negative offsets to read the ends of files, configurable line limits, exact text replacement, ambiguity detection, and directory access restrictions.

> **Deletion:** Provide a native delete operation with appropriate safeguards rather than relying on terminal commands.

2. File and content search

Capability	Description
Start a search	Search file names or file contents
More results	Retrieve paginated search results
Stop a search	Cancel an active search
List searches	View active search sessions

Search capabilities include recursive traversal, text-pattern matching, filename searching, large-result pagination, asynchronous searching, and Excel content searching.

3. Terminal and command execution

Capability	Description
Start a process	Start a command or interactive process
Read output	Retrieve new output from a running process
Send input	Send input to an existing process
Terminate a session	Stop an active terminal session
List sessions	List active terminal sessions

Other target functionality

- Bash, PowerShell, CMD, and other installed shells.
- Configurable command timeouts.
- Background and long-running execution.
- Output streaming and pagination.
- Persistent REPL sessions.
- Interactive Python, Node.js, and similar runtimes.
- Working with Git, SSH, compilers, package managers, and development tools through terminal commands.
- Capturing stdout, stderr, exit status, and process information.

> **Priority:** Terminal execution is one of the most important capabilities.

4. Process and system management

Capability	Description
List processes	List running system processes
Kill a process	Terminate processes by PID

Also manageable through terminal commands

- Linux systemd services.
- Windows services.
- Installed applications.
- Docker and Podman.
- Git repositories and developer environments.
- Scheduled tasks.
- System logs.
- Network interfaces and diagnostic utilities.

These are reached through OS commands rather than dedicated tools.

5. Document and structured-file operations

File handlers for common formats.

File type	Target functionality
TXT / Markdown	Read, create, edit, append, search
Python / JS / source code	Read, edit, search, rewrite
JSON / YAML / XML	Read and edit structured text
CSV	Read and analyze, including via terminal tools
Excel (.xlsx, .xls, .xlsm)	Read cells, write data, edit ranges, search sheets
PDF	Extract text, read pages, create and modify PDFs
Word (.docx)	Read, create, inspect, and edit content
Images	Read and present supported image formats
Web URLs	Fetch supported remote file contents

- PDF: create documents, and insert or remove pages.
- Excel: sheet selection and cell ranges.
- Word: text-bearing outlines and targeted XML editing.

6. Remote device management

Capability	Description
List devices	Display registered devices and online/offline status
Ping	Verify device connectivity
Shut down an agent	Shut down a remote agent process
Account identity	Identify the authenticated account
Remote tool routing	Execute a tool against a selected registered device
Device pairing	Register and authorize machines
Agent reconnection	Restore remote service availability after interruption

> **Self-hosted:** The relay and device agents run on infrastructure ContextBridge controls, with native agents for Linux and Windows. No third-party hosted relay.

7. Configuration and permissions

Capability	Description
Read configuration	Retrieve configuration and environment information
Change configuration	Change supported configuration values

The configuration system includes

- Allowed filesystem directories.
- Blocked commands.
- Preferred shell.
- File reading and writing limits.
- Telemetry settings.
- Connected-client information.
- Server and operating-system details.

Critical distinction: Filesystem allowlists do not automatically restrict all possible filesystem access through terminal commands. ContextBridge should enforce its security boundaries consistently across both file tools and command execution.

8. Monitoring, diagnostics, and session history

Capability	Description
Usage statistics	Retrieve usage metrics
Recent tool calls	Inspect recent tool activity
Remote call results	Retrieve pending or completed remote call results
Long-running remote calls	Inspect remote calls continuing beyond the initial wait

This covers execution history, troubleshooting, usage visibility, and asynchronous call tracking.

Audit: Preserve a durable local audit log instead of relying exclusively on in-memory session history.

9. ContextBridge implementation checklist

The following is the actionable scope, by area.

Implementation tracker — 0 / 45 complete. Check completed capabilities as they are implemented and verified. All items start unchecked; this is not a live repository status report.

Filesystem	**0/6**	**Verified (date / notes)**
☐ Read and read multiple files		
☐ Create, overwrite, and append files		
☐ Targeted file editing		
☐ Create, move, rename, and delete directories/files		
☐ Recursive listing and metadata		
☐ Path and permission enforcement		

Search	**0/6**	**Verified (date / notes)**
☐ Search filenames		
☐ Search file contents		
☐ Recursive search		
☐ Paginated results		
☐ Asynchronous search sessions		
☐ Cancel active searches		

Terminal	**0/6**	**Verified (date / notes)**
☐ Execute Bash, PowerShell, and CMD		
☐ Persistent interactive shells		
☐ Stream stdout and stderr		
☐ Send input to active processes		
☐ Timeouts and bounded output		
☐ Terminate process trees		

System and processes	**0/4**	**Verified (date / notes)**
☐ List and terminate processes		
☐ Manage authorized system services		
☐ Inspect system information and logs		
☐ Support development tools and Git		

Documents	0/5	Verified (date / notes)
☐ Excel read/write/edit/search		
☐ PDF read/create/modify		
☐ Word read/create/edit		
☐ Images and supported URLs		
☐ CSV and structured data handling		

Remote access	0/6	Verified (date / notes)
☐ Linux and Windows native agents		
☐ Device enrollment and revocation		
☐ Online/offline status and heartbeat		
☐ Encrypted bidirectional connection		
☐ Self-hosted relay with reconnect		
☐ Per-device command routing		

Security and operations	0/6	Verified (date / notes)
☐ OAuth-compatible MCP client access		
☐ Per-device and per-tool permission profiles		
☐ Credential and secret isolation		
☐ Complete audit and execution history		
☐ Configurable scopes and resource limits		
☐ Recovery after interrupted operations		

Synchronization	0/6	Verified (date / notes)
☐ Bidirectional file transfer		
☐ Directory synchronization		
☐ Checksums and change detection		
☐ Conflict preservation		
☐ Resumable transfers		
☐ Recoverable deletion propagation		

10. Synchronization beyond remote access

For this objective, there is an important difference between remote access and automatic synchronization.

Remote access lets an AI assistant read a file from **Friday**, modify it, and write a file to **Thanatos**. That is not a continuous bidirectional file synchronization engine with conflict resolution.

ContextBridge goes beyond remote access by providing native synchronization, resumable transfers, hash verification, explicit conflict handling, and recovery after network disconnections.

Recommendation: Treat sections 1–8 as the baseline scope, then add bidirectional synchronization on top of ContextBridge's own device infrastructure.

Acceptance test: ContextBridge operates against Friday and Thanatos using only its own agents, permissions, transport, and tools, with no third-party remote-access service.